

1. Questions

Read the data carefully and answer the questions based on it-

There are five different games available on a website X, i.e., A, B, C, D, and E. The number of users who like to play these games include children, teenagers, and adults. The table given below shows the average number of users and percentage of children out of the total number of users who like to play these games and also gives the percentage of adult users by which they are less or more than that of teenagers.

Games	Average number of users	Percentage of children	% of which adult users are more or less than teenagers
A	-	26%	+50%
B	360	-	-44%
C	500	18%	-
D	-	40%	+14.28%
E	-	28.57%	-

Notes:

I). Some data is missing in the table.

II). + and – signs denote the number of adults more and less than that of teenagers respectively.

III). An average number of users who play these five games on the website X is 1265, which is 130% more than the number of children who like to play game D.

IV). The difference between the total number of users who play games E and A is 130.

V). All values are integers.

Out of the total number of adults who play game A, two-thirds are males, and the total number of adult females who play game D is 245. Find the average number of males playing games A and D is what percent of the total number of users playing game A?

- 28.8%
- 32.6%
- 22.6%
- 24.8%
- None of these

@QuizByMickey

2. Questions

The ratio of the number of teenagers who play games B and E is 25:18, and the average number of children and teenagers who play game B is 400. If the difference between the number of teenagers and adults who play game E is “M” then find the number of factors of M?

- a. 8
- b. 12
- c. 6
- d. 10
- e. none of these

3. Questions

The average number of adults and teenagers who play game C is approximately what percent more or less than the total number of adults and teenagers who play game A?

- a. 27%
- b. 34%
- c. 39%
- d. 49%
- e. None of these

4. Questions

Out of the total number of children who play game D, 27.27% are below 8 years of age, and X% of the remaining are above 10 years of age. If the number of children whose ages are above 8 years and below 10 years (including 8 and 10 years of age) is 128, then find the $(X - 3)\%$ number of adults who play game D?

- a. 217
- b. 245
- c. 287
- d. 286
- e. 299

5. Questions

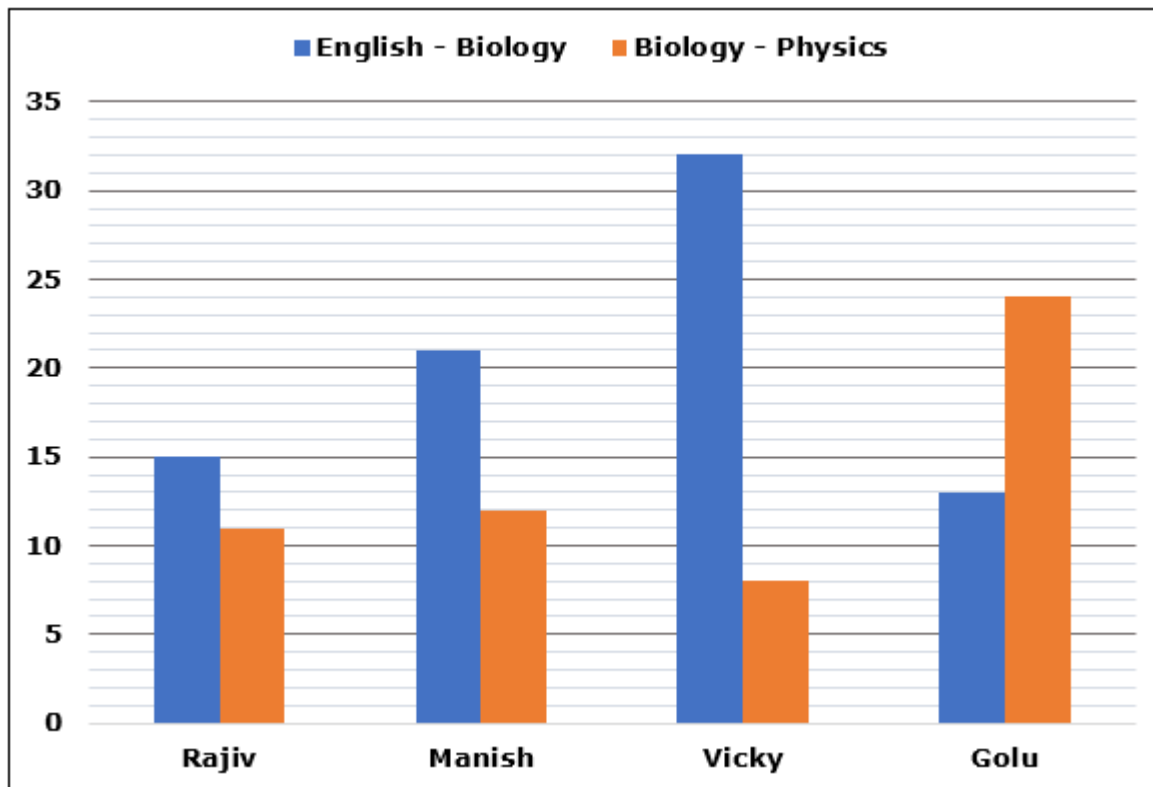
If the number of children who play game B is 107.14% of the number of adults who play the same game, then among all these five games, which game has the third highest number of children as users?

- a. Game A
- b. Game B
- c. Game C
- d. Game D
- e. Game E

6. Questions

Read the data carefully and answer the questions based on it.

The data is related to the number of questions attempted by four students Rajiv, Manish, Vicky and Golu in three different subjects i.e. English, Biology and Physics. The given bar graph shows the difference between the number of questions attempted by each student in English and biology (English – biology) and the difference between the number of questions attempted by each student in biology and physics (biology – physics). The maximum number of questions in each subject is 120, and for every correct answer, 1.5 marks are rewarded, while for every incorrect answer, there is a negative marking of 0.75 marks.



Notes:

- I). The total number of questions attempted by Manish in all three subjects is 285.
- II). No one attempted less than 60 questions in any subject.
- III). The average number of questions attempted by Vicky in all three subjects together is $(2Y+18)$, which is 37.5% more than the number of questions attempted by Rajiv in physics.
- IV). The average number of questions attempted by Golu in biology and physics together is $(6Z)$.
- V). The value of Y is the product of two consecutive single-digit prime numbers and the value of Z is also a product of two consecutive single-digit prime numbers, such that the HCF of Y and Z is greater than 1. The product of Y and Z is an odd number, and $Y > Z$.

If the ratio of wrong questions answered by Manish and Golu in physics is 2:3 and Manish scored 12 marks more than Golu in physics, then find the ratio of right questions answered by Manish and Golu in physics.

a. 12:11

- b. 8:7
- c. 9:11
- d. 11:13
- e. None of these

7. Questions

If 24 questions answered by Rajiv are wrong, then find the aggregate percentage scored by him in all three subjects together.

- a. 78.95%
- b. 56.25%
- c. 51.75%
- d. 53.61%
- e. None of these

8. Questions

Shivani attempted 14.28% less questions than Vicky in English, and the average number of questions attempted by her in biology and physics together is 5.26% less than that of Vicky. If the difference between the number of questions attempted by her in English is 25 more than that in biology, then the number of questions unattempted by her in physics?

- a. 53
- b. 47
- c. 42
- d. 38
- e. 97

9. Questions

A pass mark is 45% of the maximum marks, and if Rajiv failed by 16.5 marks in physics and Manish failed by 1.5 marks in the same subject, then find the sum of wrong questions attempted by Rajiv and Manish in physics.

- a. 36
- b. 28
- c. 32
- d. 40
- e. None of these

10. Questions

Find the difference between the total number of questions attempted by Rajiv in English and physics together and the number of questions attempted by Golu in biology and physics together.

- a. $1.2Z + 8$
- b. $0.8Y - 4$
- c. $0.4Y + 0.8Z$
- d. Both A and B
- e. Both A and C

11. Questions

Read the following data carefully and answer the questions based on it.

A: $x^2 + 2(5a)x = 18b$, where “16” and “-t” are the roots of the equations and the value of t is less than 50.

B: $(m + [2n+p])^{(2/3)} - m/3 = 3$ (where $n = 0.25t$)

C: m and n are the roots of the equation: $y^2 - 48y + 5t = -171$ and $m > n$.

Find the value of $[b+p] + [m-n]/3$.

- a. $(16a + 12)$
- b. $[(n)^2 + 2p]$
- c. p^2
- d. $(2m/13)^2$
- e. None of these

12. Questions

Which among the following equations has n and p as their roots?

- a. $x^2 - a^4 + 8n = 0$
- b. $x^2 - (b/8)^2 + 63 = 0$
- c. $x^2 - 2(p+1) + n^2 - 8 = 0$
- d. Both B and C
- e. Both A and B

@QuizByMickey

13. Questions

Which among the following options is true regarding the value of $(t+m)$?

- a. The number of factors is 5
- b. When $3a$ is added, it becomes a perfect square
- c. It is one of the factors of $[14(b+3)+45]$
- d. More than one option is correct
- e. All are incorrect

14. Questions

The average of four consecutive prime numbers is $(4p+a)$, and the difference between the smallest and largest number is 14; then, find the difference between the 3rd smallest number and “2m”?

- a. 49
- b. 51
- c. 47
- d. 41
- e. None of these

15. Questions

The value of $(2b+1.5t)$ is approximately what percent more or less than the value of “ $3p^2$ ”?

- a. 15%
- b. 28%
- c. 20%
- d. 24%
- e. 9%

16. Questions

A car can cover a distance of $(5N - 60)$ km from Bhopal to Delhi in 24 hours. If the speed of the car is increased by 60%, then it takes $(24 - M)$ hours. Find the value of $[M+N]$?

- a. 42
- b. 48
- c. 36
- d. Can't be determined
- e. None of these

17. Questions

The ratio of the cost price and selling price of an article is 8:9, and the marked price of the article

is “10Y” more than the cost price. If the selling price and cost price of the article are increased by Rs. 80 and the marked price remains the same, then the discount given will be Rs. 20 less than twice the profit earned before. Find the selling price of the article when it is sold at the profit of (Y-13)%.

Note: Value of Y is the positive root of equation $X^2 - 30X - 864 = 0$

- a. Rs. 1620
- b. Rs. 1512
- c. Rs. 1485
- d. Rs. 1215
- e. Rs. 1712

18. Questions

The time taken by boat P to cover 270 km upstream is the same as the time taken by boat Q to cover 495 km downstream. The upstream speed of boat Q is 58.33% less than the downstream speed of boat P. If the speed of the stream is 9 km/h, then find the total time taken by boats P and Q to each cover a distance of 216 km in still water.

Note: Both boats P and Q running in a same river.

- a. 17 hours
- b. 21 hours
- c. 14 hours
- d. 23 hours
- e. 13 hours

19. Questions

Two vessels A and B contain a mixture of tea and sugar in ratios of 3:4 and 5:8, respectively. 35 kg and 65 kg mixtures from vessels A and B are taken out and poured into another vessel C, respectively. Then from vessel C, 10 kg of the mixture is replaced with sugar. Find the ratio of sugar to tea in vessel C after replacement?

- a. 15:7
- b. 12:5
- c. 16:9
- d. 15:8
- e. None of these

@QuizByMickey

20. Questions

Aditya and Shekhar started a business and the investment of Aditya was Rs 900 more than that of

Shekhar. After 6 months Shubham joined the business by investing Rs 2000 more than that of Shekhar. If at the end of two years, the average profit received by Shekhar and Shubham is 14% less than the profit received by Aditya, then find the initial investment of Shekhar.

- a. Rs. 2400
- b. Rs. 3600
- c. Rs. 2500
- d. Rs. 1600
- e. Cannot be determined

Explanations:

1. Questions

Total number of users on the website = $1265 \times 5 = 6325$

Number of children who like to play game D = $1265/23 \times 10 = 550$

Total number of users who play game D = $550/2 \times 5 = 1375$

Game A

Let the total number of users = $100a$

Number of children = $26a$

Number of teenagers = $29.6a$

Number of adults = $44.4a$

Game B

Total number of users = 1080

Game C

Total number of users = 1500

Number of children = 270

Game D

Total number of users = 1375

Number of children = 550

Number of teenagers + Number of adults = $1375 - 550 = 825$

Number of teenagers : Number of adults = 7 : 8

Number of teenagers = $825/15 \times 7 = 385$

Number of adults = $825 - 385 = 440$

@QuizByMickey

Now sum of number of users who play game A and E = 2370

Difference between number of users who play game A and E = 130

Case (I)

Number of users who play game A > Number of users who play game E

Number of users who play game A = 1250

Number of users who play game E = 1120

$100a = 1250$

$a = 12.5$

Number of children = $26 \times 12.5 = 325$

Number of teenagers = $29.6 \times 12.5 = 370$

Number of adults = $44.4 \times 12.5 = 555$

Case (II)

Number of users who play game A < Number of users who play game E

Number of users who play game A = 1120

Number of users who play Game E = 1250

$100a = 1120$

$a = 11.2$

Number of children = $26 \times 11.2 = 291.2$ (not possible)

Number of children who play game E = 320

Games	Total number of users	Number of children	Number of teenagers	Number of adults
A	1250	325	370	555
B	1080	-	-	-
C	1500	270	-	-
D	1375	550	385	440
E	1120	320	-	-

@QuizByMickey

Answer: C

Number of adults who play game A = 555

Number of males who play game A = $555/3 \times 2 = 370$

Number of males who play game D = $440 - 245 = 195$

Average number of males who play game A and D together = 282.5

$$282.5/1250 \times 100 = 22.6\%$$

Hence, the answer is option C.

2. Questions

$$\text{Total number of users on the website} = 1265 \times 5 = 6325$$

$$\text{Number of children who like to play game D} = 1265/23 \times 10 = 550$$

$$\text{Total number of users who play game D} = 550/2 \times 5 = 1375$$

Game A

$$\text{Let the total number of users} = 100a$$

$$\text{Number of children} = 26a$$

$$\text{Number of teenagers} = 29.6a$$

$$\text{Number of adults} = 44.4a$$

Game B

$$\text{Total number of users} = 1080$$

Game C

$$\text{Total number of users} = 1500$$

$$\text{Number of children} = 270$$

Game D

$$\text{Total number of users} = 1375$$

$$\text{Number of children} = 550$$

$$\text{Number of teenagers} + \text{Number of adults} = 1375 - 550 = 825$$

$$\text{Number of teenagers} : \text{Number of adults} = 7 : 8$$

$$\text{Number of teenagers} = 825/15 \times 7 = 385$$

$$\text{Number of adults} = 825 - 385 = 440$$

$$\text{Now sum of number of users who play game A and E} = 2370$$

$$\text{Difference between number of users who play game A and E} = 130$$

Case (I)

$$\text{Number of users who play game A} > \text{Number of users who play game E}$$

$$\text{Number of users who play game A} = 1250$$

$$\text{Number of users who play game E} = 1120$$

$$100a = 1250$$

$$a = 12.5$$

@QuizByMickey

$$\text{Number of children} = 26 \times 12.5 = 325$$

$$\text{Number of teenagers} = 29.6 \times 12.5 = 370$$

$$\text{Number of adults} = 44.4 \times 12.5 = 555$$

Case (II)

Number of users who play game A < Number of users who play game E

$$\text{Number of users who play game A} = 1120$$

$$\text{Number of users who play Game E} = 1250$$

$$100a = 1120$$

$$a = 11.2$$

$$\text{Number of children} = 26 \times 11.2 = 291.2 \text{ (not possible)}$$

$$\text{Number of children who play game E} = 320$$

Games	Total number of users	Number of children	Number of teenagers	Number of adults
A	1250	325	370	555
B	1080	-	-	-
C	1500	270	-	-
D	1375	550	385	440
E	1120	320	-	-

Answer: D

$$\text{Total number of children and teenagers who play game B} = 800$$

$$\text{Number of adults who play game B} = 280$$

$$\text{Number of teenagers who play game B} = 280/14 \times 25 = 500$$

$$\text{Number of teenagers playing game E} = 500/25 \times 18 = 360$$

$$\text{Number of adults who play game E} = 1120 - 320 - 360 = 440$$

$$\text{Difference between number of adults and teenagers who play game E} = 440 - 360$$

$$M = 80$$

$$\text{Number of factors of 80} = 10$$

Hence, the answer is option D.

3. Questions

$$\text{Total number of users on the website} = 1265 \times 5 = 6325$$

$$\text{Number of children who like to play game D} = 1265/23 \times 10 = 550$$

Total number of users who play game D = $550/2 \times 5 = 1375$

Game A

Let the total number of users = $100a$

Number of children = $26a$

Number of teenagers = $29.6a$

Number of adults = $44.4a$

Game B

Total number of users = 1080

Game C

Total number of users = 1500

Number of children = 270

Game D

Total number of users = 1375

Number of children = 550

Number of teenagers + Number of adults = $1375 - 550 = 825$

Number of teenagers : Number of adults = 7 : 8

Number of teenagers = $825/15 \times 7 = 385$

Number of adults = $825 - 385 = 440$

Now sum of number of users who play game A and E = 2370

Difference between number of users who play game A and E = 130

Case (I)

Number of users who play game A > Number of users who play game E

Number of users who play game A = 1250

Number of users who play game E = 1120

$100a = 1250$

$a = 12.5$

Number of children = $26 \times 12.5 = 325$

Number of teenagers = $29.6 \times 12.5 = 370$

Number of adults = $44.4 \times 12.5 = 555$

Case (II)

Number of users who play game A < Number of users who play game E

@QuizByMickey

Number of users who play game A = 1120

Number of users who play Game E = 1250

$$100a = 1120$$

$$a = 11.2$$

Number of children = $26 \times 11.2 = 291.2$ (not possible)

Number of children who play game E = 320

Games	Total number of users	Number of children	Number of teenagers	Number of adults
A	1250	325	370	555
B	1080	-	-	-
C	1500	270	-	-
D	1375	550	385	440
E	1120	320	-	-

Answer: B

Average number of adults and teenagers who play game C = $1230/2 = 615$

Total number of adults and teenagers who play game A = 925

$$= (925 - 615) / 925 \times 100$$

$$= 33.51\% \sim 34\%$$

Hence, the answer is option B.

4. Questions

Total number of users on the website = $1265 \times 5 = 6325$

Number of children who like to play game D = $1265/23 \times 10 = 550$

Total number of users who play game D = $550/2 \times 5 = 1375$

Game A

Let the total number of users = $100a$

Number of children = $26a$

Number of teenagers = $29.6a$

Number of adults = $44.4a$

Game B

Total number of users = 1080

Game C

@QuizByMickey

Total number of users = 1500

Number of children = 270

Game D

Total number of users = 1375

Number of children = 550

Number of teenagers + Number of adults = $1375 - 550 = 825$

Number of teenagers : Number of adults = 7 : 8

Number of teenagers = $825/15 * 7 = 385$

Number of adults = $825 - 385 = 440$

Now sum of number of users who play game A and E = 2370

Difference between number of users who play game A and E = 130

Case (I)

Number of users who play game A > Number of users who play game E

Number of users who play game A = 1250

Number of users who play game E = 1120

$100a = 1250$

$a = 12.5$

Number of children = $26 * 12.5 = 325$

Number of teenagers = $29.6 * 12.5 = 370$

Number of adults = $44.4 * 12.5 = 555$

Case (II)

Number of users who play game A < Number of users who play game E

Number of users who play game A = 1120

Number of users who play Game E = 1250

$100a = 1120$

$a = 11.2$

Number of children = $26 * 11.2 = 291.2$ (not possible)

Number of children who play game E = 320

@QuizByMickey

Games	Total number of users	Number of children	Number of teenagers	Number of adults
A	1250	325	370	555
B	1080	-	-	-
C	1500	270	-	-
D	1375	550	385	440
E	1120	320	-	-

Answer: D

Total number of children who play game D = 550

Number of children below 8 years of age = $550 \times \frac{3}{11} = 150$

Number of children **whose ages are above 8 years and below 10 years (including 8 and 10 years of age)** = 128

Number of children above 10 years of age = 272

$272/400 \times 100 = X\%$

$68\% = X\%$

Number of adults who play game d = 440

$440 \times (68 - 3)\% = 286$

Hence, the answer is option C.

5. Questions

Total number of users on the website = $1265 \times 5 = 6325$

Number of children who like to play game D = $1265/23 \times 10 = 550$

Total number of users who play game D = $550/2 \times 5 = 1375$

Game A

Let the total number of users = 100a

Number of children = 26a

Number of teenagers = 29.6a

Number of adults = 44.4a

Game B

Total number of users = 1080

Game C

Total number of users = 1500

@QuizByMickey

Number of children = 270

Game D

Total number of users = 1375

Number of children = 550

Number of teenagers + Number of adults = $1375 - 550 = 825$

Number of teenagers : Number of adults = 7: 8

Number of teenagers = $825/15 * 7 = 385$

Number of adults = $825 - 385 = 440$

Now sum of number of users who play game A and E = 2370

Difference between number of users who play game A and E = 130

Case (I)

Number of users who play game A > Number of users who play game E

Number of users who play game A = 1250

Number of users who play game E = 1120

$100a = 1250$

$a = 12.5$

Number of children = $26 * 12.5 = 325$

Number of teenagers = $29.6 * 12.5 = 370$

Number of adults = $44.4 * 12.5 = 555$

Case (II)

Number of users who play game A < Number of users who play game E

Number of users who play game A = 1120

Number of users who play Game E = 1250

$100a = 1120$

$a = 11.2$

Number of children = $26 * 11.2 = 291.2$ (not possible)

Number of children who play game E = 320

@QuizByMickey

Games	Total number of users	Number of children	Number of teenagers	Number of adults
A	1250	325	370	555
B	1080	-	-	-
C	1500	270	-	-
D	1375	550	385	440
E	1120	320	-	-

Answer: E

Game A = 325

Game B = number of children = $280/14 \times 15 = 300$

Game C = 270

Game D = 550

Game E = 320

Hence, the answer is option B.

6. Questions

Single-digit prime number = 2,3,5,7

If $y = 3 \times 5 = 15$

Then $z = 2 \times 3 = 6$

$15 \times 6 = 90$ (not an odd number)

If $y = 5 \times 7 = 35$

Then $z = 3 \times 5 = 15$

$(35 \times 15) = \text{odd number}$

So, $y = 35$

$Z = 15$

Let E = english, P = physics and B = biology

For Manish

$E - B = 21$

$B - P = 12$

$E = 21 + B$

$P = B - 12$

Total number of questions attempted by Manish = 285

@QuizByMickey

$$21 + B + b - 12 + B = 285$$

$$3B = 276$$

$$B = 92$$

$$E = 113$$

$$P = 80$$

For Vicky

$$\text{Total number of questions attempted by Vicky} = (2 \times 35 + 18) \times 3 = 264$$

$$E - B = 32$$

$$E = 32 + B$$

$$B - P = 8$$

$$P = B - 8$$

$$3B + 24 = 264$$

$$3B = 240$$

$$B = 80$$

$$P = 72$$

$$E = 112$$

For Rajiv

$$\text{Number of questions attempted in physics} = 88/11 \times 8 = 64$$

$$B - P = 11$$

$$B = 75$$

$$E - B = 15$$

$$E = 90$$

For golu

$$\text{Number of questions attempted in bio and physics together} = 180$$

$$B - P = 24$$

$$B = 102$$

$$P = 78$$

$$E - B = 13$$

$$E = 115$$

@QuizByMickey

Students	Number of questions attempted in English	Number of questions attempted in Biology	Number of questions attempted in Physics	Total number of questions attempted
Rajiv	90	75	64	229
Manish	113	92	80	285
Vicky	112	80	72	264
Golu	115	102	78	295

Answer: A

Let the number of wrong answers marked by Manish in Physics = $2a$

Number of wrong answers marked by Golu in Physics = $3a$

$$[80 \times 1.5 - 2a \times 1.5 - (2a \times 0.75)] - [78 \times 1.5 - 3a \times 1.5 - 3a \times 0.75] = 12$$

$$120 - 4.5a - 117 + 6.75a = 12$$

$$2.25a = 9$$

$$a = 4$$

Number of right answers marked by Manish in Physics = $80 - 8 = 72$

Number of right answers marked by Golu in Physics = $78 - 12 = 66$

Required ratio = $72:66 = 12:11$

Hence, the answer is option A.

7. Questions

Single-digit prime number = 2,3,5,7

$$\text{If } y = 3 \times 5 = 15$$

$$\text{Then } z = 2 \times 3 = 6$$

$$15 \times 6 = 90 \text{ (not an odd number)}$$

$$\text{If } y = 5 \times 7 = 35$$

$$\text{Then } z = 3 \times 5 = 15$$

$$(35 \times 15) = \text{odd number}$$

$$\text{So, } y = 35$$

$$Z = 15$$

Let E = english, P = physics and B = biology

For Manish

$$E - B = 21$$

@QuizByMickey

$$B - P = 12$$

$$E = 21 + B$$

$$P = B - 12$$

Total number of questions attempted by Manish = 285

$$21 + B + B - 12 + B = 285$$

$$3B = 276$$

$$B = 92$$

$$E = 113$$

$$P = 80$$

For Vicky

Total number of questions attempted by Vicky = $(2 \times 35 + 18) \times 3 = 264$

$$E - B = 32$$

$$E = 32 + B$$

$$B - P = 8$$

$$P = B - 8$$

$$3B + 24 = 264$$

$$3B = 240$$

$$B = 80$$

$$P = 72$$

$$E = 112$$

For Rajiv

Number of questions attempted in physics = $88/11 \times 8 = 64$

$$B - P = 11$$

$$B = 75$$

$$E - B = 15$$

$$E = 90$$

For golu

Number of questions attempted in bio and physics together = 180

$$B - P = 24$$

$$B = 102$$

$$P = 78$$

@QuizByMickey

$$E - B = 13$$

$$E = 115$$

Students	Number of questions attempted in English	Number of questions attempted in Biology	Number of questions attempted in Physics	Total number of questions attempted
Rajiv	90	75	64	229
Manish	113	92	80	285
Vicky	112	80	72	264
Golu	115	102	78	295

Answer: D

Total number of questions attempted by Rajiv = 229

Marks scored by Rajiv = $229 \times 1.5 - 24 \times 1.5 - 24 \times 0.75 = 289.5$

Total marks of all the subjects = $(120 + 120 + 120) \times 1.5 = 360 \times 1.5 = 540$

Required percentage = $289.5/540 \times 100$

= 53.61%

Hence, the answer is option D.

8. Questions

Single-digit prime number = 2,3,5,7

If $y = 3 \times 5 = 15$

Then $z = 2 \times 3 = 6$

$15 \times 6 = 90$ (not an odd number)

If $y = 5 \times 7 = 35$

Then $z = 3 \times 5 = 15$

$(35 \times 15) = \text{odd number}$

So, $y = 35$

$Z = 15$

Let E = english, P = physics and B = biology

For Manish

$$E - B = 21$$

$$B - P = 12$$

$$E = 21 + B$$

@QuizByMickey

$$P = B - 12$$

Total number of questions attempted by Manish = 285

$$21 + B + b - 12 + B = 285$$

$$3B = 276$$

$$B = 92$$

$$E = 113$$

$$P = 80$$

For Vicky

Total number of questions attempted by Vicky = $(2 \times 35 + 18) \times 3 = 264$

$$E - B = 32$$

$$E = 32 + B$$

$$B - P = 8$$

$$P = B - 8$$

$$3B + 24 = 264$$

$$3B = 240$$

$$B = 80$$

$$P = 72$$

$$E = 112$$

For Rajiv

Number of questions attempted in physics = $88/11 \times 8 = 64$

$$B - P = 11$$

$$B = 75$$

$$E - B = 15$$

$$E = 90$$

For golu

Number of questions attempted in bio and physics together = 180

$$B - P = 24$$

$$B = 102$$

$$P = 78$$

$$E - B = 13$$

$$E = 115$$

@QuizByMickey

Students	Number of questions attempted in English	Number of questions attempted in Biology	Number of questions attempted in Physics	Total number of questions attempted
Rajiv	90	75	64	229
Manish	113	92	80	285
Vicky	112	80	72	264
Golu	115	102	78	295

Answer: B

Number of questions attempted by Shivani in English = $112 \times \frac{6}{7} = 96$

Average no. Of questions attempted by Vicky in bio and physics together = 76

Average no. Of questions attempted by Shivani in bio and physics together = $76 \times \frac{18}{19} = 72$

Total number of questions attempted by her in biology and physics together = 144

Number of questions attempted by her in biology = $96 - 25 = 71$

Number of questions attempted by her in physics = $144 - 71 = 73$

No. Of questions unattempt by her in physics = $120 - 73 = 47$

Hence, the answer is option B.

9. Questions

Single-digit prime number = 2,3,5,7

If $y = 3 \times 5 = 15$

Then $z = 2 \times 3 = 6$

$15 \times 6 = 90$ (not an odd number)

If $y = 5 \times 7 = 35$

Then $z = 3 \times 5 = 15$

$(35 \times 15) = \text{odd number}$

So, $y = 35$

$Z = 15$

Let E = english, P = physics and B = biology

For Manish

$E - B = 21$

$B - P = 12$

$E = 21 + B$

@QuizByMickey

$$P = B - 12$$

Total number of questions attempted by Manish = 285

$$21 + B + b - 12 + B = 285$$

$$3B = 276$$

$$B = 92$$

$$E = 113$$

$$P = 80$$

For Vicky

Total number of questions attempted by Vicky = $(2 \times 35 + 18) \times 3 = 264$

$$E - B = 32$$

$$E = 32 + B$$

$$B - P = 8$$

$$P = B - 8$$

$$3B + 24 = 264$$

$$3B = 240$$

$$B = 80$$

$$P = 72$$

$$E = 112$$

For Rajiv

Number of questions attempted in physics = $88/11 \times 8 = 64$

$$B - P = 11$$

$$B = 75$$

$$E - B = 15$$

$$E = 90$$

For golu

Number of questions attempted in bio and physics together = 180

$$B - P = 24$$

$$B = 102$$

$$P = 78$$

$$E - B = 13$$

$$E = 115$$

@QuizByMickey

Students	Number of questions attempted in English	Number of questions attempted in Biology	Number of questions attempted in Physics	Total number of questions attempted
Rajiv	90	75	64	229
Manish	113	92	80	285
Vicky	112	80	72	264
Golu	115	102	78	295

Answer: C

Maximum marks = $120 \times 1.5 = 180$

Passing marks = $180 \times 45\% = 81$

Let wrong questions attempted by Rajiv in physics = x

$$64 \times 1.5 - x \times 1.5 - 0.75x = 64.5$$

$$96 - 2.25x = 64.5$$

$$2.25x = 31.5$$

$$X = 14$$

Let the wrong questions attempted by Manish in physics = y

$$80 \times 1.5 - y \times 1.5 - 0.75y = 79.5$$

$$120 - 2.25y = 79.5$$

$$2.25y = 40.5$$

$$Y = 18$$

$$x + y = 32$$

Hence, the answer is option C.

10. Questions

Single-digit prime number = 2,3,5,7

$$\text{If } y = 3 \times 5 = 15$$

$$\text{Then } z = 2 \times 3 = 6$$

$$15 \times 6 = 90 \text{ (not an odd number)}$$

$$\text{If } y = 5 \times 7 = 35$$

$$\text{Then } z = 3 \times 5 = 15$$

$$(35 \times 15) = \text{odd number}$$

$$\text{So, } y = 35$$

@QuizByMickey

$$Z = 15$$

Let E = english, P = physics and B = biology

For Manish

$$E - B = 21$$

$$B - P = 12$$

$$E = 21 + B$$

$$P = B - 12$$

Total number of questions attempted by Manish = 285

$$21 + B + B - 12 + B = 285$$

$$3B = 276$$

$$B = 92$$

$$E = 113$$

$$P = 80$$

For Vicky

Total number of questions attempted by Vicky = $(2 \times 35 + 18) \times 3 = 264$

$$E - B = 32$$

$$E = 32 + B$$

$$B - P = 8$$

$$P = B - 8$$

$$3B + 24 = 264$$

$$3B = 240$$

$$B = 80$$

$$P = 72$$

$$E = 112$$

For Rajiv

Number of questions attempted in physics = $88/11 \times 8 = 64$

$$B - P = 11$$

$$B = 75$$

$$E - B = 15$$

$$E = 90$$

For golu

@QuizByMickey

Number of questions attempted in bio and physics together =180

$$B - P = 24$$

$$B = 102$$

$$P = 78$$

$$E - B = 13$$

$$E = 115$$

Students	Number of questions attempted in English	Number of questions attempted in Biology	Number of questions attempted in Physics	Total number of questions attempted
Rajiv	90	75	64	229
Manish	113	92	80	285
Vicky	112	80	72	264
Golu	115	102	78	295

Answer: E

Number of questions attempted by Rajiv in English and physics together = 90+64 = 154

Number of questions attempted by Golu in biology and physics together = 180 = 180-154 = 26

a). $1.2Z + 8 = 26$

c). $0.4Y + 0.8Z = 26$

Hence, the answer is option E

11. Questions

$$n = 0.25t$$

$$y^2 - 48y + 5t = -171$$

n is one of the roots of the above equation and $n = 0.25t$

$$\text{So } (0.25t)^2 - 48(0.25t) + 5t + 171 = 0$$

$$t^2 - 192t + 80t + 171 \cdot 16 = 0$$

$$t^2 - 112t + 2736 = 0$$

$$t^2 - 36t - 76t + 2736 = 0$$

$$t(t - 36) - 76(t - 36) = 0$$

$$(t - 36)(t - 76) = 0$$

@QuizByMickey

$$t = 36 \text{ or } 76$$

The value of t is less than 50, so $t = 36$

$$n = 0.325 * 36 = 9$$

$$y^2 - 48y + 5 * 36 + 171 = 0$$

$$y^2 - 48y + 351 = 0$$

$$y^2 - 39y - 9y + 351 = 0$$

$$(y - 39)(y - 9) = 0$$

$$y = 39 \text{ or } 9$$

$$m = 39$$

$$B: (m + [2n + p])^{2/3} - m/3 = 3$$

$$(39 + 18 + p)^{2/3} = 16$$

$$(57 + p) = 4^3$$

$$57 + p = 64$$

$$p = 7$$

$$A: x^2 + 10ax = 18b$$

$$256 + 160a = 18b$$

$$18b - 160a = 256 \text{---(I)}$$

$$1296 - 360a = 18b$$

$$18b + 360a = 1296 \text{---(II)}$$

from equations I and II

$$520a = 1040$$

$$a = 2$$

$$18b = 1296 - 720$$

$$18b = 576$$

$$b = 32$$

$$a = 2, b = 32, t = 36, m = 39, n = 9, p = 7$$

Answer: C

$$[b + p] + [m - n]/3$$

$$= 39 + (39 - 9)/3$$

@QuizByMickey

$$= 49$$

$$= p^2$$

Hence, the answer is option (C)

12. Questions

$$n = 0.25t$$

$$y^2 - 48y + 5t = -171$$

n is one of the roots of the above equation and $n = 0.25t$

$$\text{So } (0.25t)^2 - 48(0.25t) + 5t + 171 = 0$$

$$t^2 - 192t + 80t + 171 \cdot 16 = 0$$

$$t^2 - 112t + 2736 = 0$$

$$t^2 - 36t - 76t + 2736 = 0$$

$$t(t - 36) - 76(t - 36) = 0$$

$$(t - 36)(t - 76) = 0$$

$$t = 36 \text{ or } 76$$

The value of t is less than 50, so $t = 36$

$$n = 0.25 \cdot 36 = 9$$

$$y^2 - 48y + 5 \cdot 36 + 171 = 0$$

$$y^2 - 48y + 351 = 0$$

$$y^2 - 39y - 9y + 351 = 0$$

$$(y - 39)(y - 9) = 0$$

$$y = 39 \text{ or } 9$$

$$m = 39$$

$$B: (m + [2n + p])^{2/3} - m/3 = 3$$

$$(39 + 18 + p)^{2/3} = 16$$

$$(57 + p) = 4^3$$

$$57 + p = 64$$

$$p = 7$$

$$A: x^2 + 10ax = 18b$$

@QuizByMickey

$$256 + 160a = 18b$$

$$18b - 160a = 256 \text{---(I)}$$

$$1296 - 360a = 18b$$

$$18b + 360a = 1296 \text{---(II)}$$

from equations I and II

$$520a = 1040$$

$$a = 2$$

$$18b = 1296 - 720$$

$$18b = 576$$

$$b = 32$$

$$a = 2, b = 32, t = 36, m = 39, n = 9, p = 7$$

Answer: B

$$n = 9, p = 7$$

$$\text{equation} = x^2 - 16x + 63 = 0$$

$$= x^2 - (b/8)^2 + 63 = 0$$

Hence, the answer is option B.

13. Questions

$$n = 0.25t$$

$$y^2 - 48y + 5t = -171$$

n is one of the roots of the above equation and $n = 0.25t$

$$\text{So } (0.25t)^2 - 48(0.25t) + 5t + 171 = 0$$

$$t^2 - 192t + 80t + 171 \cdot 16 = 0$$

$$t^2 - 112t + 2736 = 0$$

$$t^2 - 36t - 76t + 2736 = 0$$

$$t(t - 36) - 76(t - 36) = 0$$

$$(t - 36)(t - 76) = 0$$

$$t = 36 \text{ or } 76$$

The value of t is less than 50, so $t = 36$

$$n = 0.25 \cdot 36 = 9$$

@QuizByMickey

$$y^2 - 48y + 5 * 36 + 171 = 0$$

$$y^2 - 48y + 351 = 0$$

$$y^2 - 39y - 9y + 351 = 0$$

$$(y - 39)(y - 9) = 0$$

$$y = 39 \text{ or } 9$$

$$m = 39$$

$$B: (m + [2n+p])^{2/3} - m/3 = 3$$

$$(39+18 + p)^{2/3} = 16$$

$$(57+p) = 4^3$$

$$57 + p = 64$$

$$p = 7$$

$$A: x^2 + 10ax = 18b$$

$$256 + 160a = 18b$$

$$18b - 160a = 256 \text{---(I)}$$

$$1296 - 360a = 18b$$

$$18b + 360a = 1296 \text{---(II)}$$

from equations I and II

$$520a = 1040$$

$$a = 2$$

$$18b = 1296 - 720$$

$$18b = 576$$

$$b = 32$$

$$a = 2, b = 32, t = 36, m = 39, n = 9, p = 7$$

Answer: B

$$t + m = 36 + 39 = 75$$

A). 75 has 6 factors, so the option is incorrect.

B). $75 + 3 * 2 = 81$ (True)

C). $(14 * 35 + 45) = 535$ which is not divisible by 75, so it is incorrect

Hence, the answer is option B.

@QuizByMickey

14. Questions

$$n = 0.25t$$

$$y^2 - 48y + 5t = -171$$

n is one of the roots of the above equation and $n = 0.25t$

$$\text{So } (0.25t)^2 - 48(0.25t) + 5t + 171 = 0$$

$$t^2 - 192t + 80t + 171 \times 16 = 0$$

$$t^2 - 112t + 2736 = 0$$

$$t^2 - 36t - 76t + 2736 = 0$$

$$t(t - 36) - 76(t - 36) = 0$$

$$(t - 36)(t - 76) = 0$$

$$t = 36 \text{ or } 76$$

The value of t is less than 50, so $t = 36$

$$n = 0.25 \times 36 = 9$$

$$y^2 - 48y + 5 \times 36 + 171 = 0$$

$$y^2 - 48y + 351 = 0$$

$$y^2 - 39y - 9y + 351 = 0$$

$$(y - 39)(y - 9) = 0$$

$$y = 39 \text{ or } 9$$

$$m = 39$$

$$\text{B: } (m + [2n + p])^{2/3} - m/3 = 3$$

$$(39 + 18 + p)^{2/3} = 16$$

$$(57 + p) = 4^3$$

$$57 + p = 64$$

$$p = 7$$

$$\text{A: } x^2 + 10ax = 18b$$

$$256 + 160a = 18b$$

$$18b - 160a = 256 \text{---(I)}$$

$$1296 - 360a = 18b$$

@QuizByMickey

$$18b + 360a = 1296 \text{---(II)}$$

from equations I and II

$$520a = 1040$$

$$a = 2$$

$$18b = 1296 - 720$$

$$18b = 576$$

$$b = 32$$

$$a = 2, b = 32, t = 36, m = 39, n = 9, p = 7$$

Answer: C

$$\text{Average} = 4 \times 7 + 2 = 30$$

$$\text{Sum of all four numbers} = 120$$

$$\text{series} = 23, 29, 31, 37$$

$$\text{Third smallest number} = 31$$

$$2m = 78$$

$$78 - 31 = 47$$

Hence, the answer is option D.

15. Questions

$$n = 0.25t$$

$$y^2 - 48y + 5t = -171$$

n is one of the roots of the above equation and $n = 0.25t$

$$\text{So } (0.25t)^2 - 48(0.25t) + 5t + 171 = 0$$

$$t^2 - 192t + 80t + 171 \times 16 = 0$$

$$t^2 - 112t + 2736 = 0$$

$$t^2 - 36t - 76t + 2736 = 0$$

$$t(t - 36) - 76(t - 36) = 0$$

$$(t - 36)(t - 76) = 0$$

$$t = 36 \text{ or } 76$$

The value of t is less than 50, so $t = 36$

$$n = 0.25 \times 36 = 9$$

$$y^2 - 48y + 5 \times 36 + 171 = 0$$

@QuizByMickey

$$y^2 - 48y + 351 = 0$$

$$y^2 - 39y - 9y + 351 = 0$$

$$(y - 39)(y - 9) = 0$$

$$y = 39 \text{ or } 9$$

$$m = 39$$

$$B: (m + [2n + p])^{2/3} - m/3 = 3$$

$$(39 + 18 + p)^{2/3} = 16$$

$$(57 + p) = 4^3$$

$$57 + p = 64$$

$$p = 7$$

$$A: x^2 + 10ax = 18b$$

$$256 + 160a = 18b$$

$$18b - 160a = 256 \text{---(I)}$$

$$1296 - 360a = 18b$$

$$18b + 360a = 1296 \text{---(II)}$$

from equations I and II

$$520a = 1040$$

$$a = 2$$

$$18b = 1296 - 720$$

$$18b = 576$$

$$b = 32$$

$$a = 2, b = 32, t = 36, m = 39, n = 9, p = 7$$

Answer: C

$$2b + 1.5t = 2 \times 32 + 1.5 \times 36 = 118$$

$$3p^2 = 3 \times 49 = 147$$

$$(147 - 118) / 147 \times 100$$

$$19.72\% \sim 20\%$$

Hence, the answer is option C.

16. Questions

@QuizByMickey

Answer: D

Let the speed of Car = X kmph

$$24 * X = 5N - 60$$

If the Speed of a car increased by 60%

$$(24 - M) * X * 8/5 = 5N - 60$$

$$24X = (24 - M) * X * 8/5$$

$$15 = 24 - M$$

$$M = 9$$

The value of N can't be determined

Hence, the answer is option D.

17. Questions

Answer: B

Let the cost price = Rs. 8a

Selling price = Rs. 9a

Marked price = Rs. [8a + 10Y]

$$X^2 - 30X - 864 = 0$$

$$X^2 - 48X + 18x - 864 = 0$$

$$X = +48 \text{ or } -18$$

$$Y = 48$$

Marked price = 8a + 480

If selling price = 9a + 80

Then discount = 400 - a

$$400 - a = 2a - 20$$

$$3a = 420$$

$$a = 140$$

Cost price of the article = Rs 1120

Selling price when it sold at the profit of 35%

$$1120 * 135\%$$

$$= \text{Rs. } 1512$$

Hence, the answer is option B.

18. Questions

@QuizByMickey

Answer: A

Speed of stream = 9kmph

Let the downstream speed of boat P = $12x$

Upstream speed of boat Q = $5x$

Speed of boat P in still water = $12x - 9$

Speed of boat Q in still water = $5x + 9$

According to question

$$270/(12x - 9) = 495/(5x + 9)$$

$$17x = 51$$

$$x = 3$$

Speed of boat P in still water = 27 kmph

Speed of boat Q in still water = 24 kmph

Required time = $216/27 + 216/24$

$$= 8+9$$

$$= 17 \text{ hours}$$

Hence, the answer is option A.

19. Questions

Answer: C

Tea and sugar taken out from vessel A = $[35/7 * 3]$ and $[35/7 * 4] = 15 \text{ kg}$ and 20 kg

Tea and sugar taken out from vessel B = $[65/13 * 5]$ and $[65/13 * 8] = 25 \text{ kg}$ and 40 kg

In vessel C

$$\text{Tea} = 15 + 25 = 40 \text{ kg}$$

$$\text{Sugar} = 20 + 40 = 60 \text{ kg}$$

$$\text{Ratio of Tea and Sugar} = 40: 60 = 2: 3$$

$$\text{After replacement, the quantity of tea} = 40 - 10/5 * 2 = 36 \text{ kg}$$

$$\text{Quantity of sugar} = 60 - 10/5 * 3 + 10 = 64 \text{ kg}$$

$$\text{Required ratio} = 36:64$$

$$9: 16$$

Hence, the answer is option C.

20. Questions

Answer: D

@QuizByMickey

Let the investment of Shekhar = Rs. X

Investment of Aditya = Rs. [X +900]

Investment of Shubham = Rs. [X+2000]

Ratio of profit of Aditya, Shekhar and Shubham at the end of 2 years

$$[X+900]*2 : [X*2] : [X+2000]*1.5$$

$$[2X+1800] : [2X] : [1.5X +3000]$$

$$2(2X+1800)/(3.5X+3000) = 50/43$$

$$172X +154800 = 175X+150000$$

$$4800 = 3X$$

$$X =1600$$

Investment of Shekhar = Rs. 1600

Hence the answer is option D.

@QuizByMickey